

Chapter 3:

Continuous Random Variables

X -r.v.

$S_X = [-5, 5] \subset \mathbb{R} \rightarrow$ X -continuous r.v.
the range of X

CDF and PDF of a continuous r.v.

Def: X -r.v.

$$\text{CDF: } F_X(x) = P(X \leq x)$$

Prop:

1. $\lim_{x \rightarrow \infty} F_X(x) = 1; \lim_{x \rightarrow -\infty} F_X(x) = 0$

2. $F \nearrow$

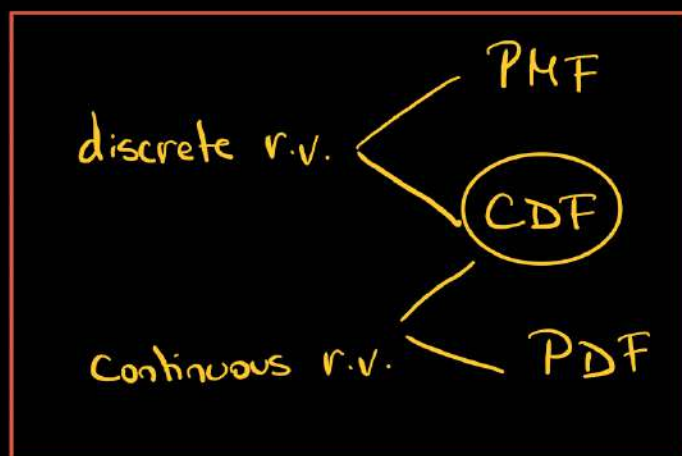
3. $P(a < X \leq b) = P_X(b) - P_X(a)$

4. $P(X = x) = 0$ (the probability of any individual outcome is 0!)

Def: X -continuous r.v. iff $F_X(x)$ is a continuous function.

Def: The probability density function PDF of a continuous r.v. is the function:

$$f_X(x) = F'_X(x), \forall x \in \mathbb{R}$$



Prop:

1. $f_X(x) \geq 0, \forall x \in \mathbb{R}$

2. $\int_{-\infty}^{\infty} f_X(x) dx = 1 \rightarrow$ that if the integral is convergent

$$\int_a^{\infty} f_X(x) dx = \lim_{t \rightarrow \infty} \int_a^t f_X(x) dx$$

$$\int_{-\infty}^a f_X(x) dx = \lim_{t \rightarrow -\infty} \int_t^a f_X(x) dx$$

$$F_X(x) \xleftrightarrow{\int} f_X(x)$$

3. $F_X(x) = \int_{-\infty}^x f_X(t) dt$

4. $P(a < X \leq b) = \int_a^b f_X(x) dx$

Ex: X -r.v. has PDF $f_X(x) = \begin{cases} c x e^{-\frac{x}{a}}, & x > 0 \\ \text{zero,} & \text{otherwise} \end{cases}$

Find:

a. $c = ?$

b. CDF = ? , $F_X(x) = ?$

c. $P(0 \leq X \leq 4) = ?$

d. $P(-2 < X < 2) = ?$

a. $f_X(x) \geq 0, \forall x \in \mathbb{R} \Rightarrow \underline{c \geq 0}$

$$\int_{-\infty}^{+\infty} f_X(x) dx = 1$$

$$1 = \int_{-\infty}^{+\infty} f_X(x) dx = \int_{-\infty}^0 0 dx + \int_0^{+\infty} c \cdot x \cdot e^{-\frac{x}{2}} dx$$

we have to compute this

$$f = x \Rightarrow f' = 1$$

$$g' = e^{-\frac{x}{2}} \Rightarrow g = \int e^{-\frac{x}{2}} dx = \frac{e^{-\frac{x}{2}}}{-\frac{1}{2}} = -2e^{-\frac{x}{2}}$$

$$\Rightarrow c \left[-2xe^{-\frac{x}{2}} \Big|_0^{+\infty} + 2 \int_0^{+\infty} e^{-\frac{x}{2}} dx \right] =$$

$$= c \left[\lim_{x \rightarrow +\infty} (-2xe^{-\frac{x}{2}}) + 0 - 4e^{-\frac{x}{2}} \Big|_0^{+\infty} \right] =$$

$$= -4c \left[\lim_{x \rightarrow +\infty} e^{-\frac{x}{2}} - e^0 \right] = 4c \Rightarrow \boxed{c = \frac{1}{4}}$$

$$\Rightarrow f_X(x) = \begin{cases} \frac{1}{4} x e^{-\frac{x}{2}}, & x \geq 0 \\ \text{zero, otherwise} \end{cases}$$

$$b. F_X(x) = \int_{-\infty}^x f_X(t) dt$$

$$\bullet \underline{x \leq 0} \Rightarrow F_X(x) = \int_{-\infty}^x 0 dt = 0$$

$$\begin{aligned} \bullet \underline{x > 0} \Rightarrow F_X(x) &= \int_{-\infty}^0 0 dt + \int_0^x \frac{1}{4} t \cdot e^{-\frac{t}{2}} dt = \\ &= \frac{1}{4} \left[-2te^{-\frac{t}{2}} \Big|_0^x - 4e^{-\frac{t}{2}} \Big|_0^x \right] = \\ &= \frac{1}{4} \left[-2xe^{-\frac{x}{2}} - 4e^{-\frac{x}{2}} + 4 \right] = \\ &= -\frac{1}{2} xe^{-\frac{x}{2}} - e^{-\frac{x}{2}} + 1 \end{aligned}$$

$$\Rightarrow F_X(x) = \begin{cases} 0, & x \leq 0 \\ -\frac{1}{2} xe^{-\frac{x}{2}} - e^{-\frac{x}{2}} + 1, & x > 0 \end{cases}$$

$$c. P(0 \leq X \leq 4) = P(0 < X \leq 4) + \underbrace{P(X=0)}_{=0} \begin{cases} F_X(4) - F_X(0) \\ \int_0^4 f_X(x) dx \end{cases}$$

$\downarrow$
 this generates problems $\Rightarrow$ let's write it in another way

$$\begin{aligned} \Rightarrow P(0 \leq X \leq 4) &= F_X(4) - F_X(0) = -\frac{1}{2} \cdot 4 \cdot e^{-\frac{4}{2}} - e^{-\frac{4}{2}} + 1 - 0 = \\ &= 1 - 3e^{-2} = \boxed{\frac{e^2 - 3}{e^2} > 0} \end{aligned}$$

$$d. P(-2 < X < 2) = P(-2 < X \leq 2) - \underbrace{P(X=2)}_{=0} =$$

$$= \int_{-2}^2 f_X(x) dx = \int_{-2}^0 0 dx + \int_0^2 \frac{1}{4} x e^{-\frac{x}{2}} dx$$

$$\text{OR } P(-2 < X < 2) = \underbrace{F_X(2) - F_X(-2)}_{=0} = -\frac{1}{2} \cdot 2e^{-1} - e^{-1} + 1 = \boxed{1 - \frac{2}{e}}$$

Expected Values

Def: The expected value of a continuous r.v. X is the number (if exists):

$$E(X) = \int_{-\infty}^{+\infty} x \cdot f_X(x) dx$$

The expected value of the r.v. $g(X)$, g -continuous:

$$E(g(X)) = \int_{-\infty}^{+\infty} g(x) \cdot f_X(x) dx$$

Prop:

$$1. E(X - E(X)) = 0$$

$$2. E(aX + b) = aE(X) + b, \quad \forall a, b \in \mathbb{R}$$

$$3. \text{The variance of } X: \text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{where } E(X^2) = \int_{-\infty}^{+\infty} x^2 \cdot f_X(x) dx$$

$$4. \text{Var}(aX + b) = a^2 \text{Var}(X)$$

Ex: PDF of X is $f_X(x) = \begin{cases} \frac{3x^2}{2}, & x \in [-1, 1] \\ \text{zero,} & \text{otherwise} \end{cases}$

a. $E(X) = ?$

b. $E(X^2) = ?$

c. $\text{Var}(X) = ?$

d. $\sigma_X = ?$

e. $F_X(x) = ?$

a. $E(X) = \int_{-\infty}^{+\infty} x \cdot f_X(x) dx \Rightarrow$

$$\Rightarrow \underline{E(X)} = \int_{-\infty}^{-1} 0 dx + \int_{-1}^1 \frac{3x^2}{2} dx + \int_1^{+\infty} 0 dx = \frac{3}{2} \cdot \frac{x^3}{3} \Big|_{-1}^1 = \frac{3}{2} (1 - (-1)) = \boxed{0}$$

b. $\underline{E(X^2)} = \int_{-\infty}^{+\infty} x^2 \cdot f_X(x) dx = \int_{-\infty}^{-1} x^2 \cdot 0 dx + \int_{-1}^1 x^2 \cdot \frac{3x^2}{2} dx + \int_1^{+\infty} x^2 \cdot 0 dx =$

$$= \frac{3}{2} \cdot \frac{x^5}{5} \Big|_{-1}^1 = \frac{3}{10} \cdot 2 = \boxed{\frac{3}{5}}$$

e. $F_X(x) = \int_{-\infty}^x f_X(t) dt$

• $x < -1$ $\Rightarrow F_X(x) = \int_{-\infty}^x 0 dt = 0$

• $x \in [-1, 1]$ $\Rightarrow F_X(x) = \int_{-\infty}^{-1} 0 dt + \int_{-1}^x \frac{3t^2}{2} dt = \frac{t^3}{2} \Big|_{-1}^x = \frac{x^3 + 1}{2}$

• $x > 1$ $\Rightarrow F_X(x) = \int_{-\infty}^{-1} 0 dt + \int_{-1}^1 \frac{3t^2}{2} dt + \int_1^x 0 dt = \frac{t^3}{2} \Big|_{-1}^1 = \frac{1}{2} + \frac{1}{2} = 1$

Families of Continuous r.v.

- ① X - uniform over $[a, b)$ if PDF is $f_X(x) = \begin{cases} \frac{1}{b-a}, & x \in [a, b) \\ \text{zero}, & \text{otherwise} \end{cases}$

Prop:

$$1. F_X(x) = \begin{cases} \text{zero}, & x \leq a \\ \frac{x-a}{b-a}, & x \in [a, b) \\ 1, & x > b \end{cases}$$

$$2. E(X) = \frac{a+b}{2}$$

$$3. \text{Var}(X) = \frac{(b-a)^2}{12}$$

- ② X - exponential r.v. of $P_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ \text{zero}, & \text{otherwise} \end{cases}, \lambda > 0$

Prop:

$$1. F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ \text{zero}, & \text{otherwise} \end{cases}$$

$$2. E(X) = \frac{1}{\lambda}$$

$$3. \text{Var}(X) = \frac{1}{\lambda^2}$$

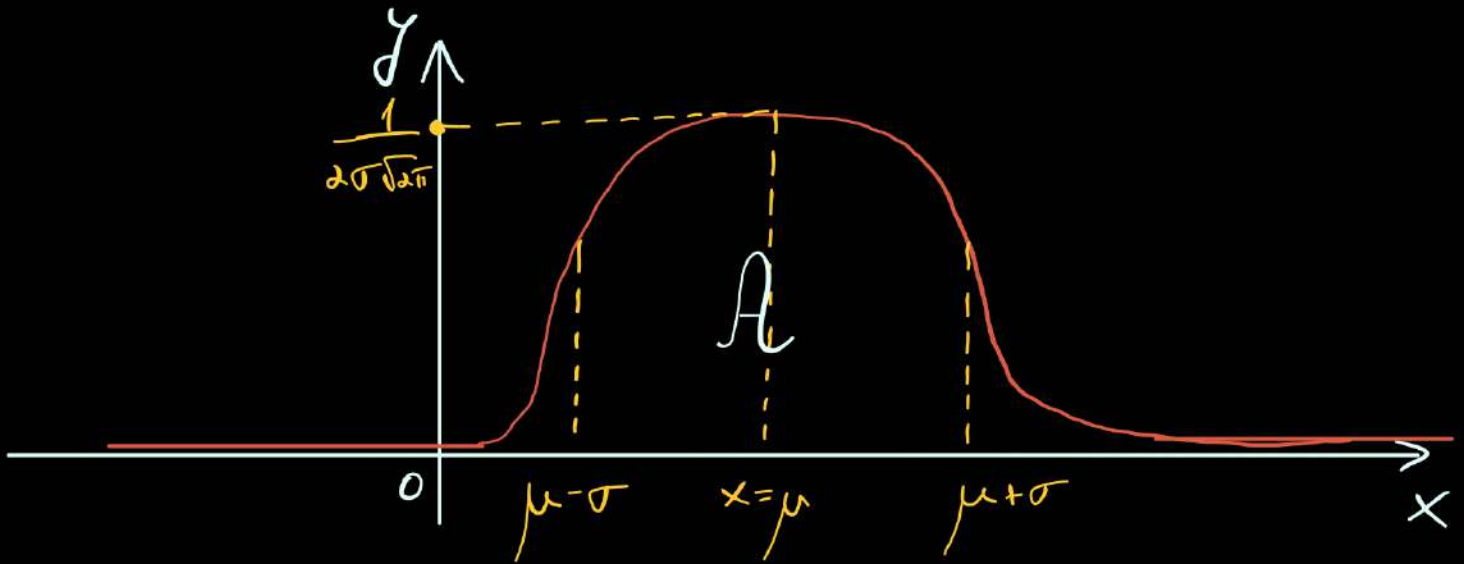
- ③ X - Gaussian (μ, σ) r.v. if PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad \begin{matrix} \mu \in \mathbb{R} \\ \sigma > 0 \end{matrix}$$

OR X - Normal r.v. $X \sim N(\mu, \sigma^2)$

Prop: $E(x) = \mu$, $\text{Var}(x) = \sigma^2$

The graph $\rightarrow$ Gauss's Bell



[0x - asymptote
 $x = \mu$ - max point
 $\mu - \sigma$, $\mu + \sigma$ - inflex points

[68% $A \rightarrow [\mu - \sigma, \mu + \sigma]$
 95% $A \rightarrow [\mu - 2\sigma, \mu + 2\sigma]$
 99,7% $A \rightarrow [\mu - 3\sigma, \mu + 3\sigma]$